

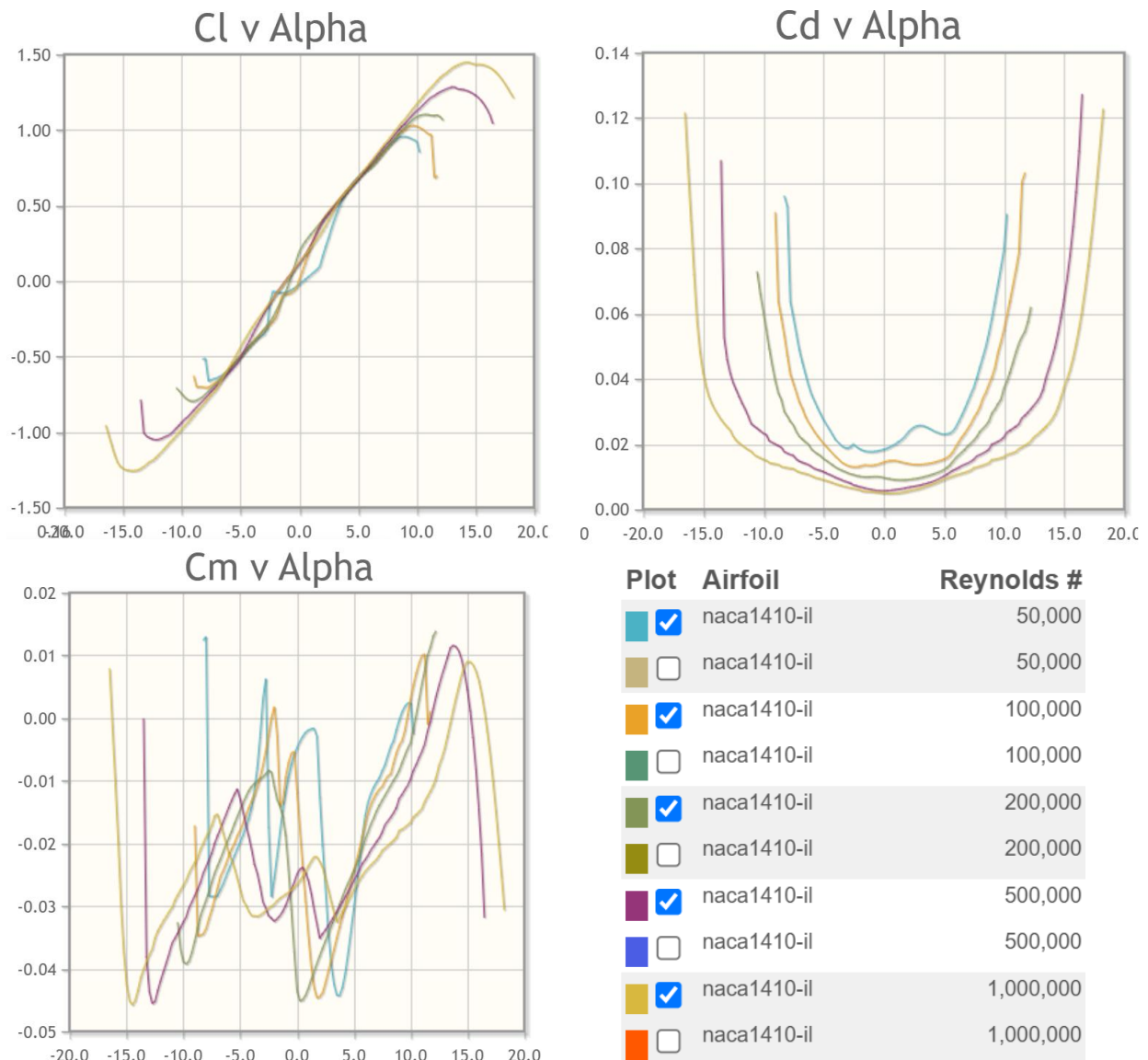
Project 1: Airfoil Analysis

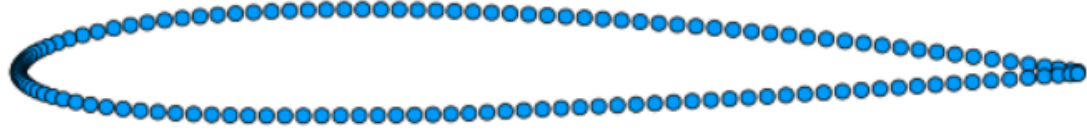
Introduction

In this paper I am going to use Xfoil to predict the behavior of different airfoils, and how different Reynolds numbers affect their coefficient of lift, drag and moment graphs. Then I am going to compare that data to experimental or other data available on online databases.

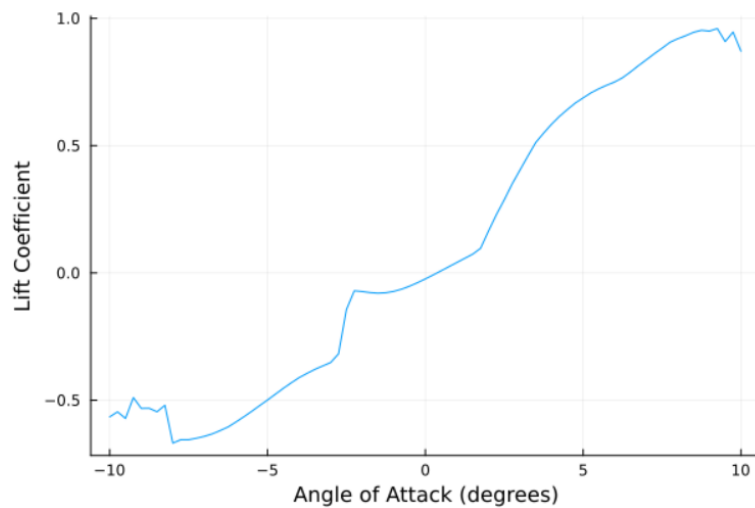
I am going to first quote experimental or other data available, followed by the data I obtained from Xfoil:

NACA 1410 reference plots

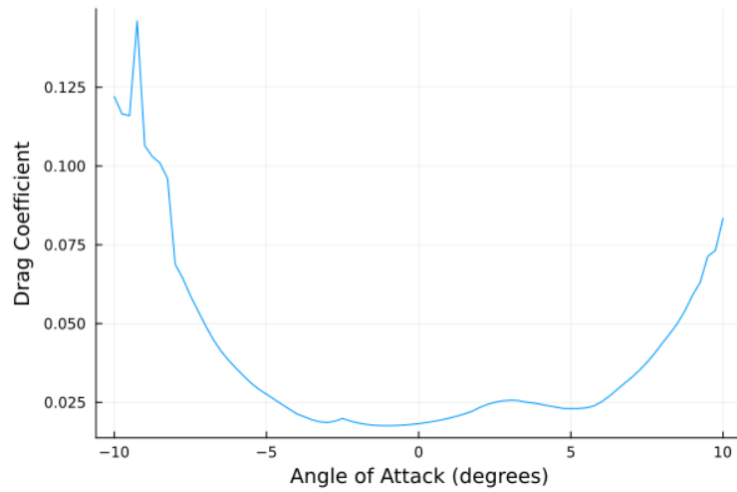




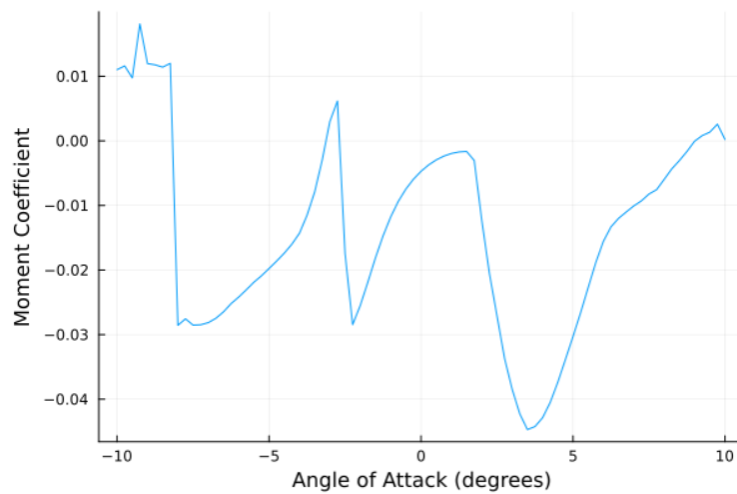
Reynolds number: 50,000



lift coefficient shows a non smooth behavior before and after -10 and 10 degrees, but a similar behavior to the one predicted by airfoil tools within this range. Their prediction also ranges from about -0.6 to 1.0

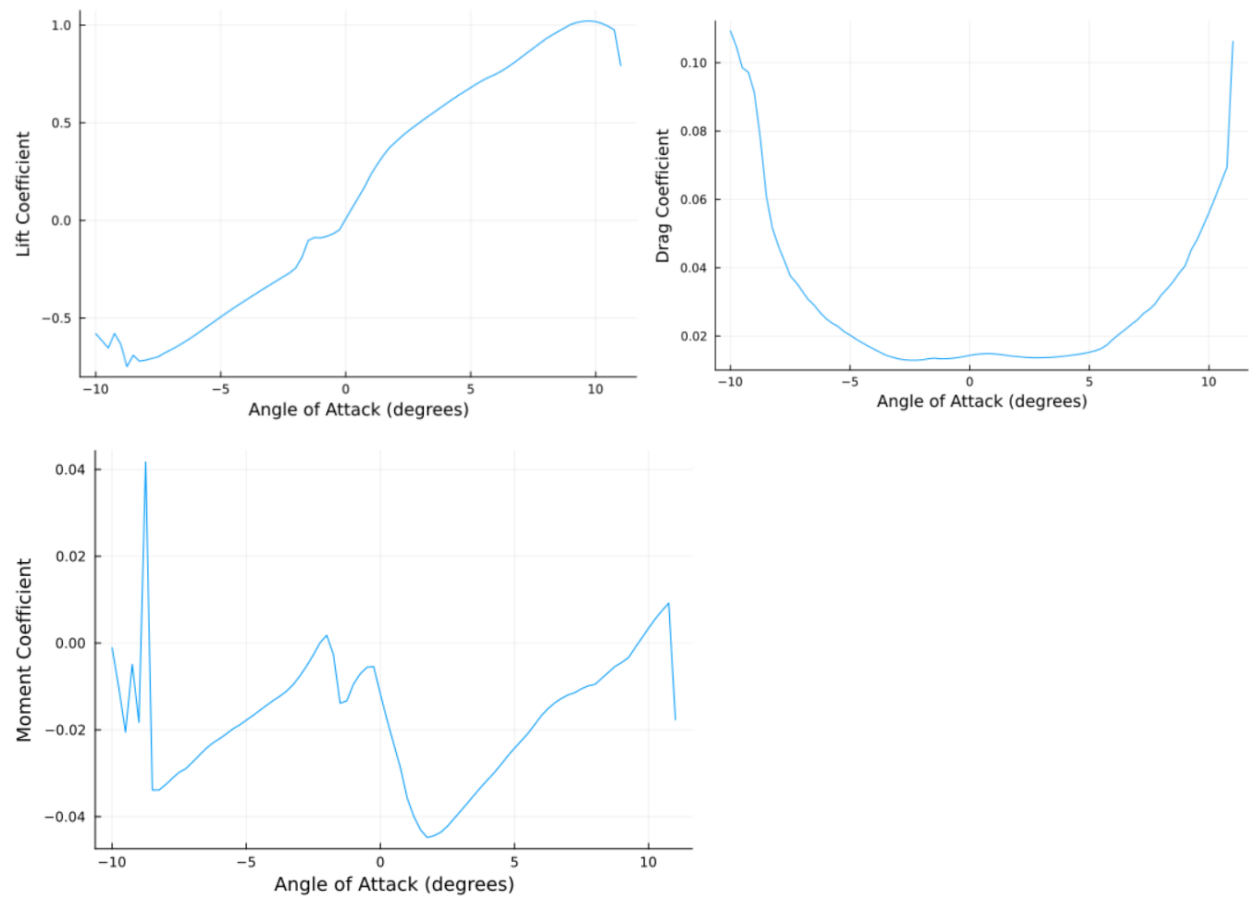


Drag has a almost identical behavior between -10 and 10 degrees, however, the behavior outside of that range is very turbulent. The reference plot does not have continuity outside of that range.

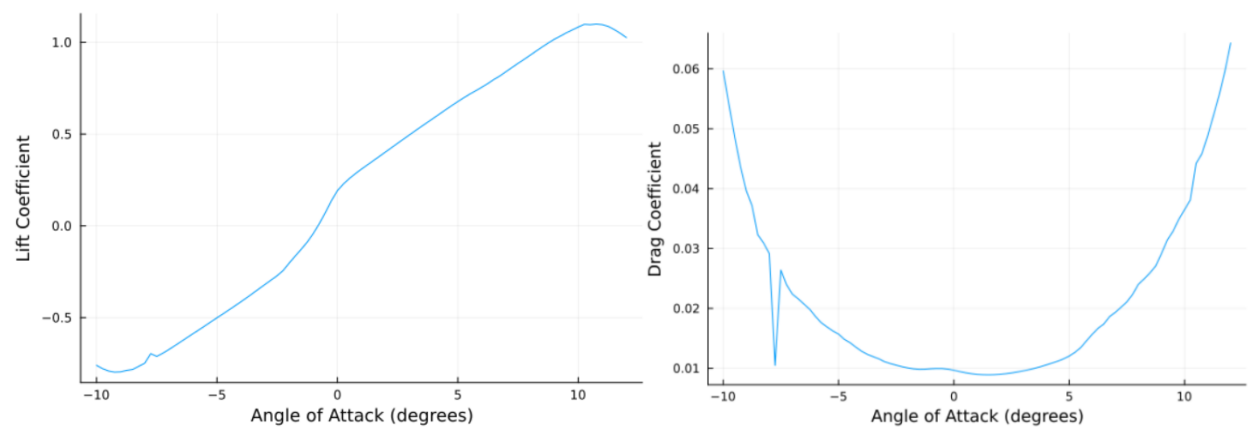


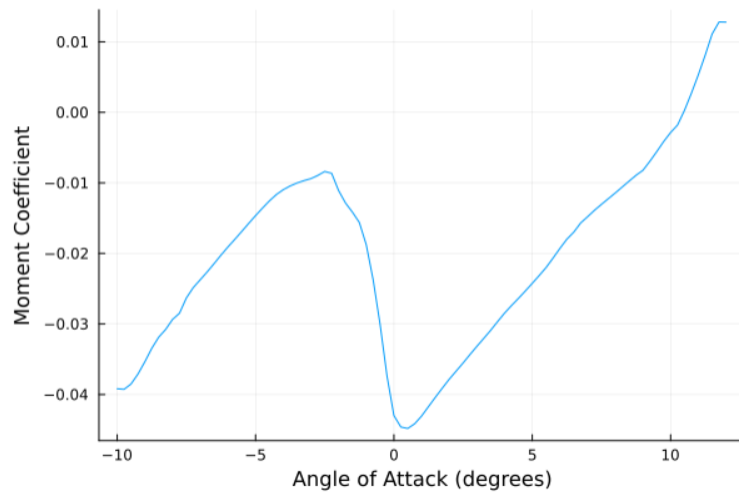
Once more, extremely similar behavior between -10, and 10, and high turbulence outside of that.

Reynolds number: 100,000

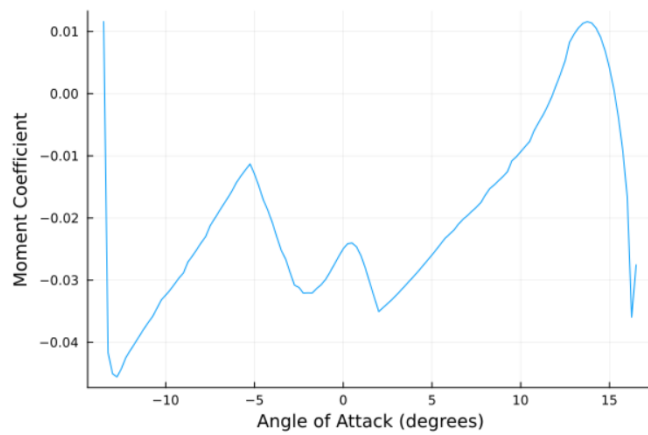
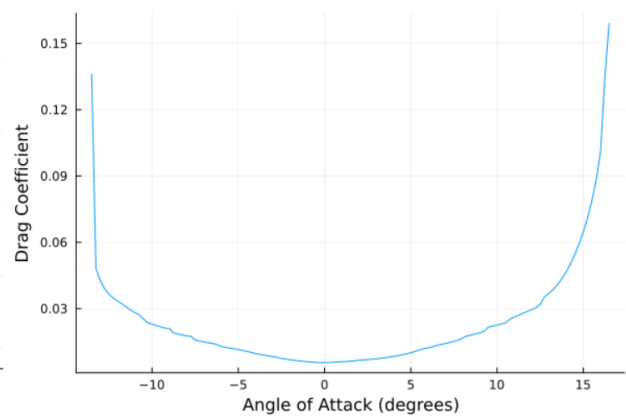
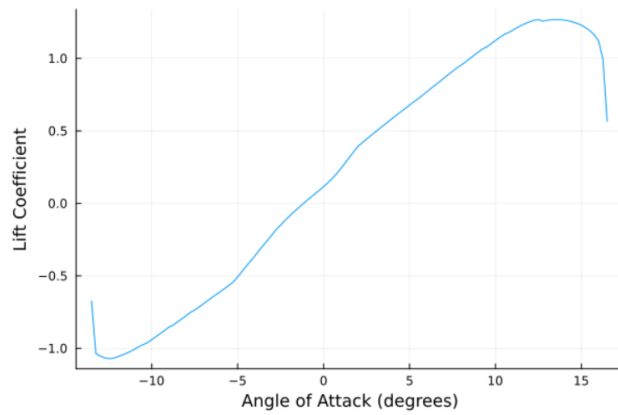


Reynolds number: 200,000

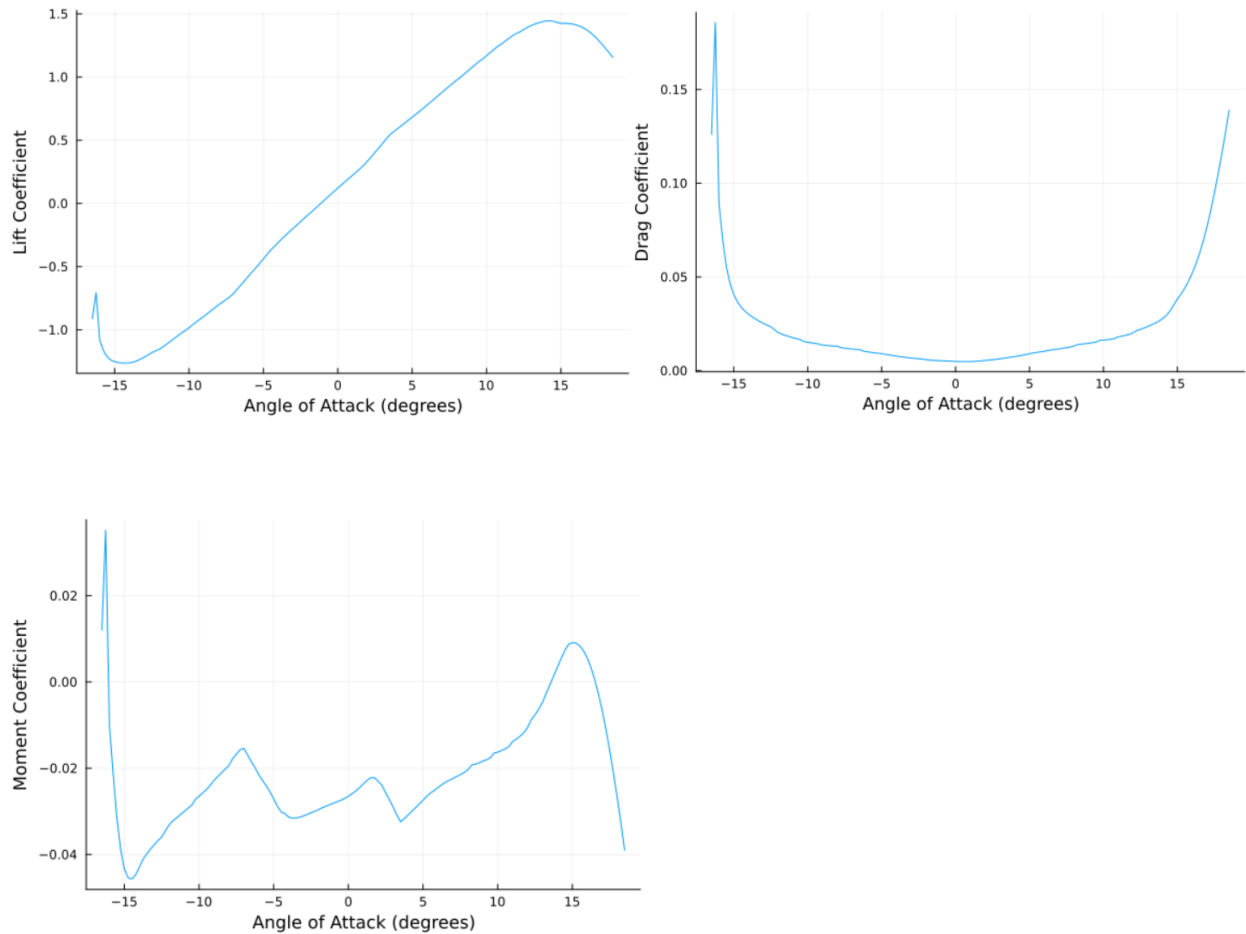




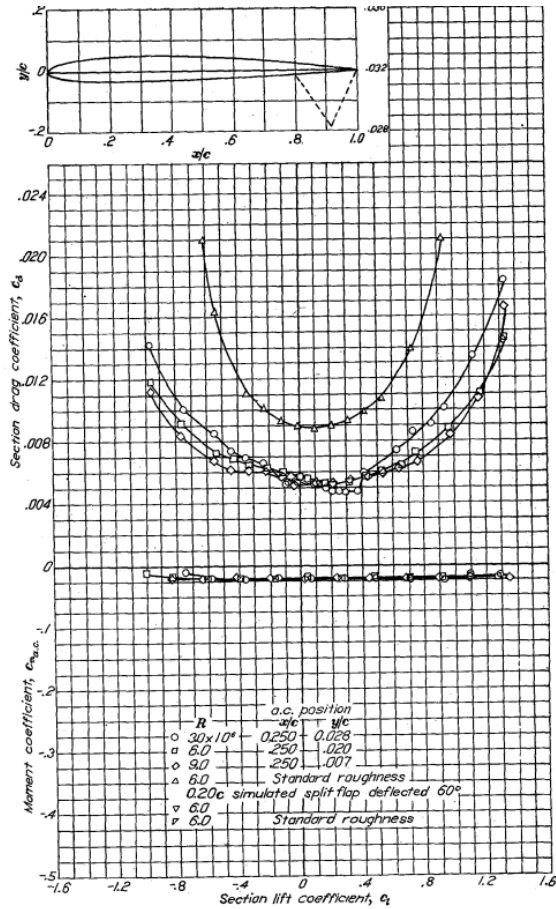
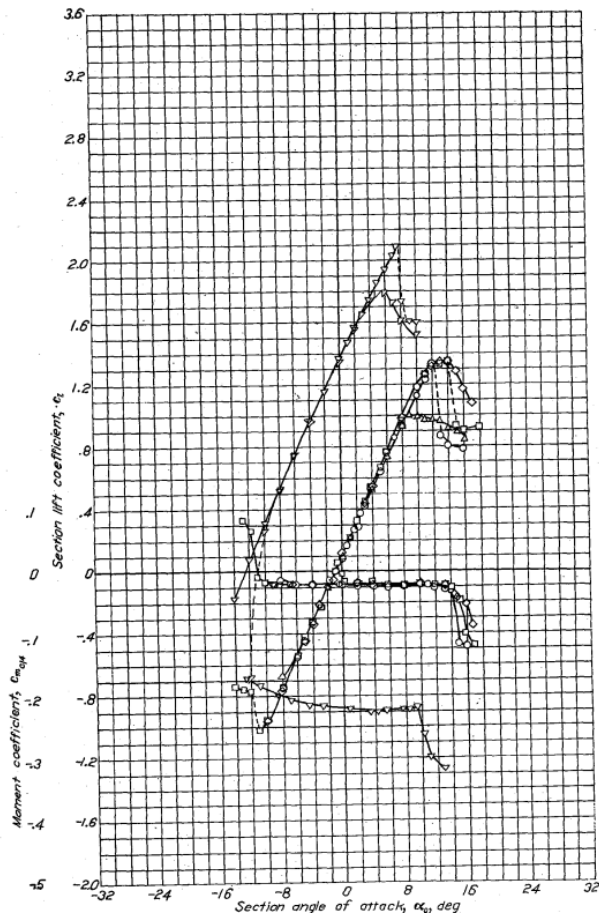
Reynolds number: 500,000



Reynolds number: 1,000,000

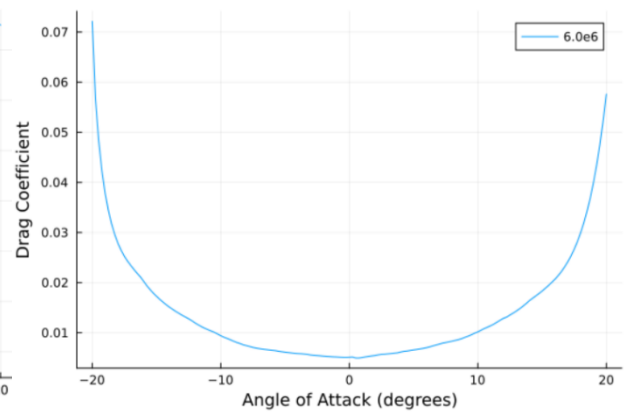
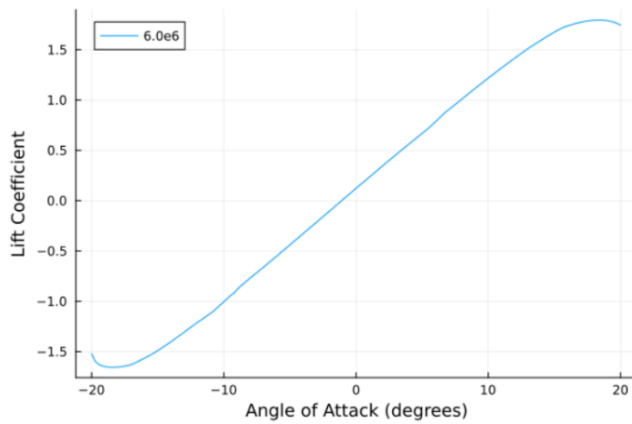


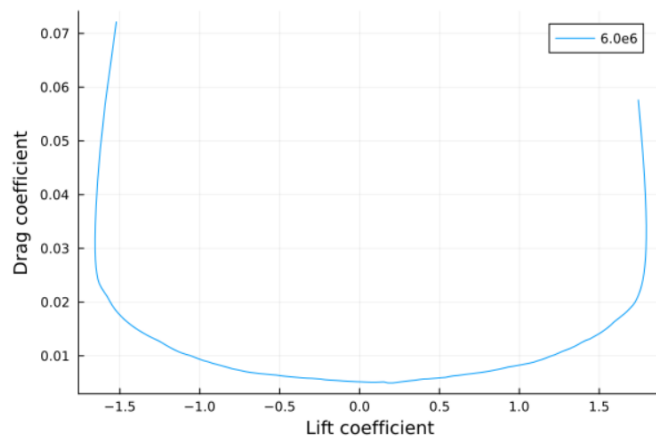
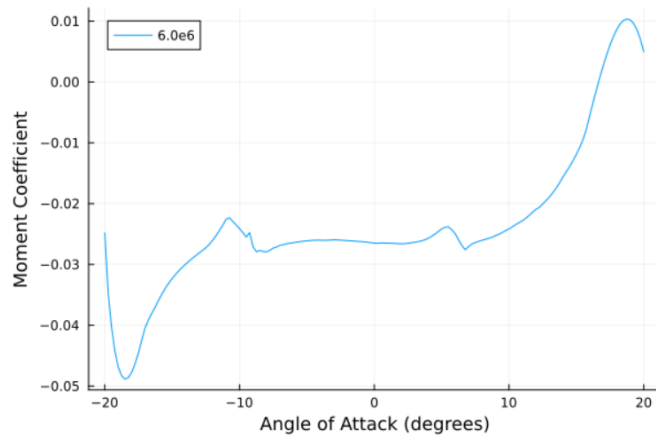
The data gathered from my code yields very similar results to that from airfoil tools. It suggests that airfoil stability has a wider range of angles of attack as Reynolds number increases. Let's now compare experimental data gathered by NASA.



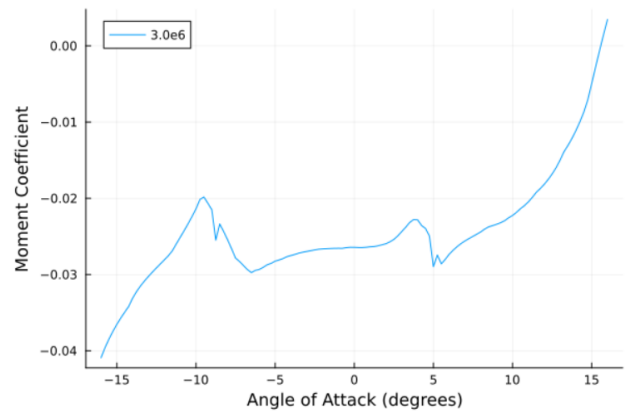
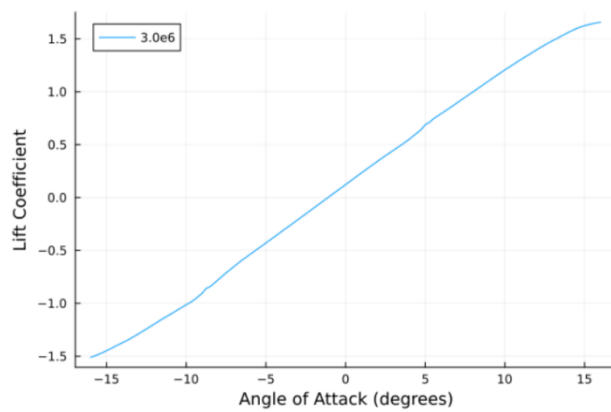
NACA 1408

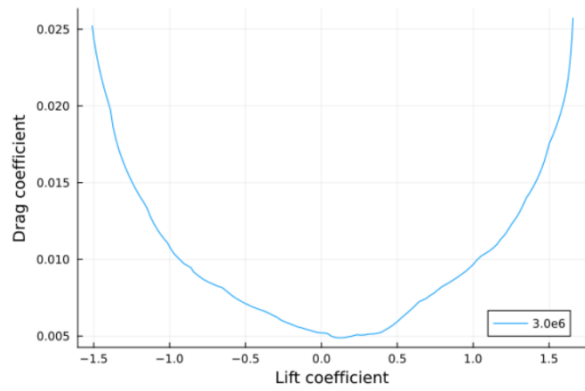
Reynolds number: 6,000,000





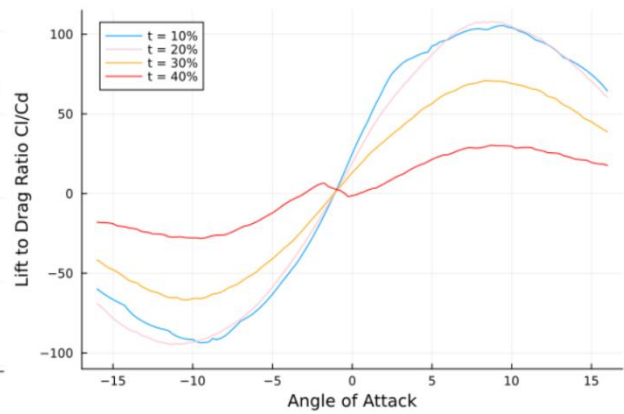
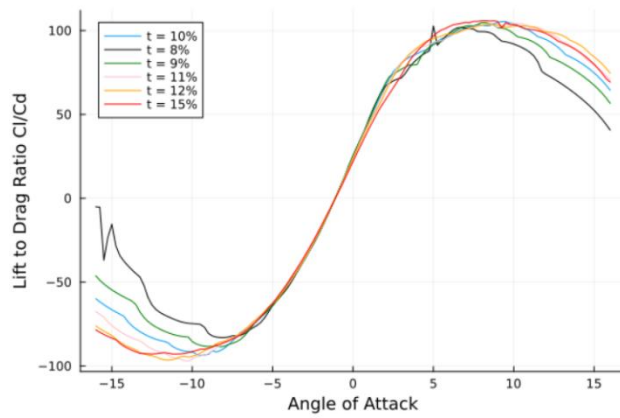
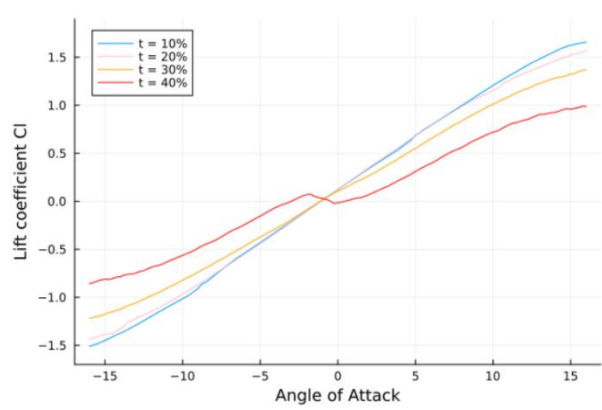
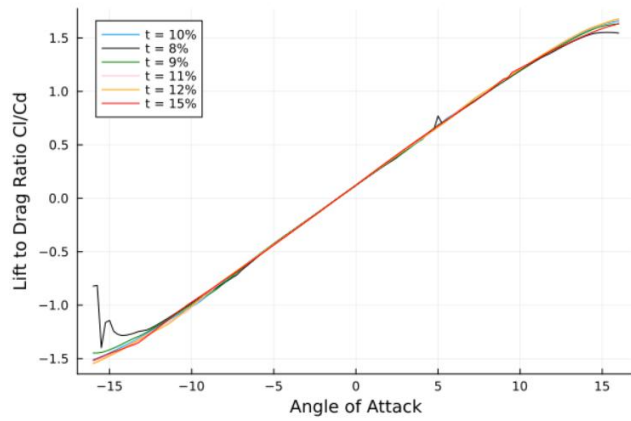
Reynolds number: 3,000,000



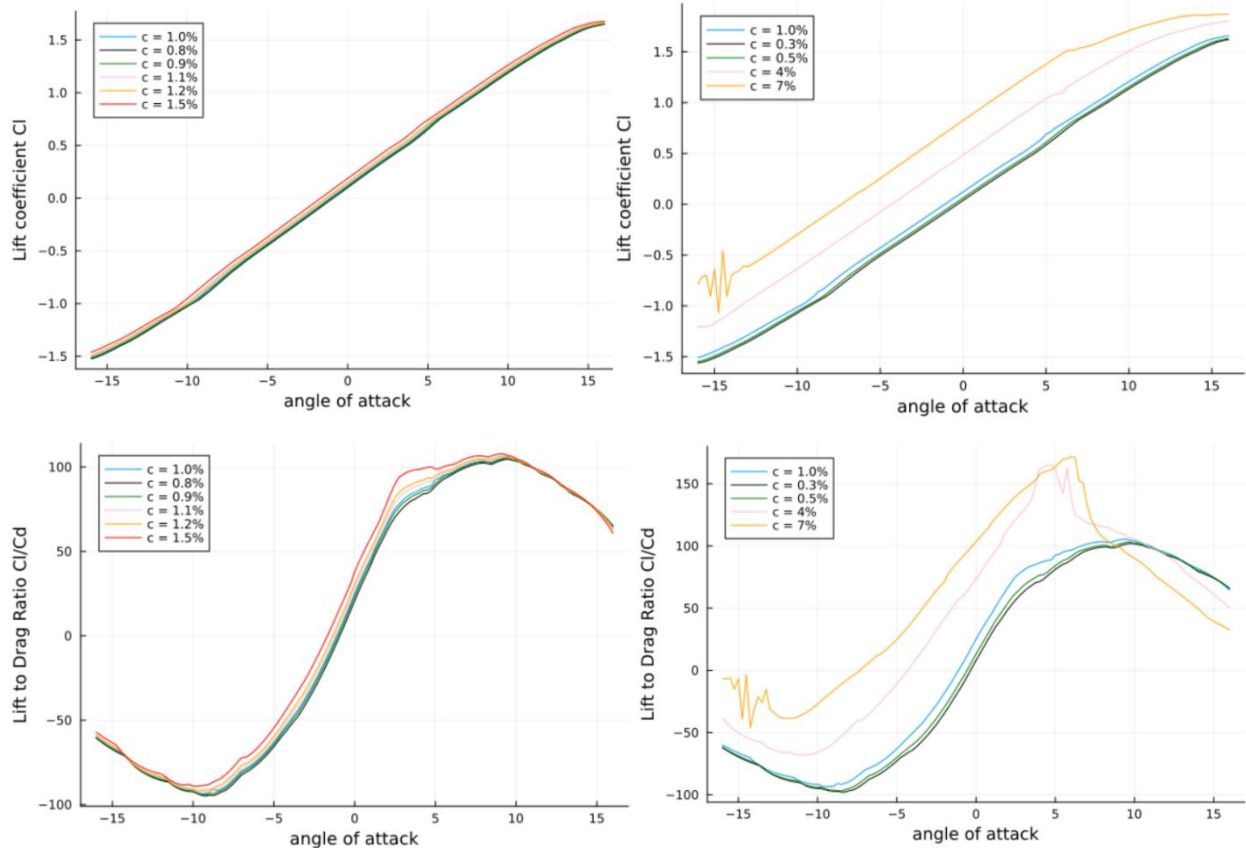


In the case of this Reynolds number, values are much closer to experimental data.

Varying thickness:



Varying max Camber line



Methods:

Airfoil:

I used Xfoil to predict the behavior of airfoil NACA 1410, which has an effective max camber of 0.97% of its chord.

Reference data:

Xfoil coefficients calculated by airfoiltools.com and experimental data compiled by NASA on the NACA 1410

Coding methods:

I used the Xfoil tool created by Mark Drela to analyze this airfoil using its *pane* and *solve_alpha* functions. Then plotted them using the Plots package. I used the same Reynolds numbers used in experimental and airfoiltools.com data. It is important to note that Xfoil's predictions were significantly close to the experimental.

Varying thicknesses:

For the varying thickness method, I started with variations of no more than 30%, however these did not show significant differences on the coefficient of lift graph. Conversely, the Coefficient to drag ratio graph did show differences before negative 10 and after positive 10 degrees. Because of these small variations I decided to use varying thicknesses with differences of over 400%

I learned how to use xfoil and **for** with variables with tuples inside to run the function I created to vary the thickness, and to plot such results

Varying camber line:

To alter the camber line, I had to:

1. Calculate the coordinates of the mean camber line
2. Find its maximum value
3. Use the max value to create a ratio with the new value I wanted to use
4. Multiply each coordinate of the original camber line by this ratio
5. Recalculate each coordinate of the new airfoil with altered camber line

Just as I did with varying thickness, I used **for** iterations with variables containing.

Results

Xfoil predictions of the base airfoil coordinates (my method) were very close to experimental data provided by NASA, with minimum variations, as the graphs show. Notice that the experimental data uses higher Reynolds' number than airfoiltools.com, hence I did different graphs comparing each.

As expected, my Xfoil predictions match airfoiltools.com because they also used Xfoil.

The most interesting part comes when varying thickness and max camber separately. The original thickness of the NACA 1410 is 10%. I tested different thicknesses keeping the camber line in place, not changing thickness by more than 1.5 times its original value. In these experiments, varying thickness did not have a significant impact on lift, but it did on drag. The results suggest that the thicker the airfoil the greater the drag only for lower values than -10 and higher than 10 degrees angle of attack.

Due to the small differences in thickness, I decided to vary it by 4 times its original value at

most. Results indicate that lift increases significantly for negative angles of attack but decreases significantly for positive angles of attack.

I ran the same procedure with camber line. Kept the original thickness the same, varying the max value for camber line keeping it at the same distance from the leading edge. Results were like those of the original airfoil. I then decided to increase the variations by at most 7 times its original max value. The data suggests that, at a greater max value for camber line, lift is greater as well.

Final words

My takeaway is what I learned calculating all the values needed for this research activity. The use of **for** loops and variables with tuples, and the use of such a useful tool as Xfoil, as well as using the Plots package effectively.